Personal Project Ideas

Minor AI for Society

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Introduction

For the personal project we have to include AI, which can be done in many different ways. At the beginning of the minor it was difficult to immediately determine which project would be the most suitable and realistic to execute within the available time.

During a feedback meeting with my teacher, Li Li (Fontys University of Applied Sciences), we discussed my current knowledge level and possible directions for a personal AI project. Based on this conversation, I decided that before committing to a specific project idea, it would be beneficial to strengthen my foundational understanding of machine learning.

For this reason, I started the Machine Learning Specialization by Andrew Ng. The goal of following this course is to establish a clearer baseline of what I already understand about AI and machine learning, as well as what skills I still need to develop. By building this foundation first, I will be better able to define a realistic and technically meaningful project later in the minor.

While I have not yet committed to one specific project, I have brainstormed several possible directions. These ideas help explore different applications of AI and allow for comparison in terms of feasibility, technical requirements, and learning potential. As I progress further in the course, I will refine these ideas and determine which one aligns best with my skills and interests.

The Ideas

As of now, I have three main project ideas that I am considering. Each of these projects incorporates AI in a different way and has its own unique focus and challenges. After completing the Machine Learning Specialization course, I will continue on working on these ideas, refining them, or maybe even coming up with new ones based on the knowledge I have gained. But for now, these three ideas serve as a starting point for exploration and comparison.

AI-Assisted Cinematic Tribute Editing Pipeline

This project extends a custom wrapper around `yt-dlp` into an AI-powered preprocessing tool for movie and TV tribute editing.

The system will automatically download selected videos and extract audio for analysis. Using speech-to-text and sentiment analysis, it will detect emotionally intense dialogue segments. Audio signal processing techniques will identify dramatic peaks and transitions.

A scoring engine will rank scenes based on emotional and cinematic weight. The tool will automatically generate timestamped highlights and export ready-to-edit clips. It will also extract impactful quotes that could be used for subtitle overlays.

All processing would run locally to ensure speed and privacy. The main goal is to reduce the amount of manual scrubbing required when creating tribute videos, while also enhancing the creative editing workflow.

This project would demonstrate applied AI techniques including natural language processing, audio signal processing, and automated media pipeline design.

Local AI-Powered Semantic File Finder

This project focuses on building a lightweight local AI tool that allows users to search their documents based on meaning instead of filenames.

The system would scan a user-selected folder and extract text from PDFs, Word documents, Markdown files, and TXT files. Documents would be split into smaller chunks (for example paragraph-level) and converted into embeddings using a small pre-trained transformer model.

These embeddings would be stored in a local vector database to enable efficient semantic search. Users could enter natural language queries to find relevant documents or passages.

The system would rank results based on semantic similarity and optionally combine this with keyword matching. A simple command line interface would display the file name, a short snippet of the relevant text, and a relevance score.

The goal is to create a practical AI-assisted document search tool that can quickly retrieve hard-to-find information. This project would demonstrate applications of AI in natural language processing, vector search, and lightweight system design.

AI-Powered Smart Desk Plant Monitor with Embedded Dashboard

This project explores the combination of embedded systems and AI by developing a smart desk plant monitoring system.

Each plant would be equipped with sensors that measure soil moisture, light intensity, temperature, and humidity. These sensors would report data to a central microcontroller that stores the readings in a local database.

A small touchscreen dashboard would provide an overview of all monitored plants, displaying color-coded health indicators. Selecting a specific plant would show detailed sensor readings, historical trends, and AI-generated care suggestions.

A TinyML model could be used to predict plant stress or watering needs based on patterns in the sensor data. Optionally, a small camera could allow AI-based plant classification to tailor care recommendations.

All processing would run locally without reliance on cloud services. The goal is to build a demonstrable and practical system that combines embedded hardware, sensor data analysis, and lightweight AI inference.

This project would demonstrate embedded system design, data collection, TinyML inference, and interactive interface development.

Next Steps

At this stage, these ideas serve as a starting point for exploration rather than finalized project proposals. As I continue progressing through the Machine Learning Specialization course, I will gain a clearer understanding of which techniques are realistic to implement and which project direction aligns best with my skills and learning goals.

Once a stronger technical baseline is established, I will refine one of these ideas into a concrete project plan with clearly defined scope, tools, and objectives.

Defining Technical Baseline & Goals

The next step in this process is to define a clear technical baseline and set specific goals for the project.

Directions in AI and Machine Learning

In general, it could be said that there are seven major directions in AI and Machine Learning that I could explore for this project.

1. Classical Machine Learning : algorithm that learn patterns from structured date.
2. Deep Learning : neural networks that can learn from unstructured data like images, text, and audio.
3. Natural Language Processing (NLP) : techniques for understanding and generating human language.
4. Computer Vision : techniques for analyzing and interpreting visual data.
5. Reinforcement Learning : learning trough trial and error by interacting with an environment.
6. Edge AI and TinyML : running AI models on low-power devices without cloud connectivity.
7. AI systems / MLOps : designing and managing machine learning models.

Core Knowledge Areas in AI

There are multiple core areas of knowdlege, which are important when it comes to AI. These domains are math, programming, machine learning concepts, data skills, and model skills.

1. Math : linear algebra, calculus, probability, and statistics.
2. Programming : proficiency in languages like Python, and familiarity with AI libraries such as TensorFlow or PyTorch.
3. Machine Learning Concepts : understanding of supervised, unsupervised, and reinforcement learning, as well as model evaluation and selection.
4. Data Skills : ability to preprocess, clean, and analyze data effectively.
5. Model Skills : experience with training, fine-tuning, and deploying machine learning models.

Generally speaking, I would say that I have a basic understanding of all of these areas, but there is still a lot to learn and improve on. Coming from ICT & Technology, I have a strong programming background and some experience with machine learning concepts. In previous projects there has been made use of computer vision, object detection, for example. Besides that, with the help of my first minor, embedded signals, I have experience with digital signal processing. Yet while I have a good foundation, I recognize that there are gaps in my knowledge, particularly in areas like deep learning and reinforcement learning. Simply as I previusly did not get the chance to dive deeper into these topics.

Directions for the Project

Based upon the ideas I brainstormed prior to defining my technical baseline and directions, I can already see that the first two ideas (AI-Assisted Cinematic Tribute Editing Pipeline and Local AI-Powered Semantic File Finder) would primarily involve natural language processing and deep learning techniques. The third idea (AI-Powered Smart Desk Plant Monitor) would involve a combination of embedded systems, sensor data analysis, and TinyML. As I would like to continue my focus on AI and embedded systems, the third idea seems to be the most aligned with my interests and goals for this project. It combines Edge AI and TinyML with practical applications in a way that is both demonstrable and engaging. The given project could be adapated in such a way it could become a general health monitoring platform, which also does align with the fact that in the future I would love to develop something that people with fore example disabilties could profit form, but for the demonstration purposes, I think it would be more interesting to focus on a specific use case, such as desk plants.

Concrete Project Proposal

The project will involve designing and building an embedded system that monitors the health of desk plants using various sensors. The system will collect data on soil moisture, light intensity, temperature, and humidity, and store this data in a local database. A small touchscreen dashboard will provide an overview of all monitored plants, displaying health indicators and detailed sensor readings. A TinyML model will be developed to analyze the sensor data and provide care suggestions based on patterns that indicate plant stress or watering needs. Optionally, a camera could be integrated to allow for AI-based plant classification, which would further tailor care recommendations. All processing will run locally to ensure privacy and low latency, making it a practical tool for desk plant management. This project will demonstrate skills in embedded system design, sensor data analysis, TinyML inference, and interactive interface development, while also providing a tangible and useful application of AI technology.

Conclusion

In conclusion, the process of brainstorming and defining project ideas has been valuable in exploring different applications of AI and understanding the technical requirements for each. And choosing for the AI-Powered Smart Desk Plant Monitor with Embedded Dashboard as the most promising direction as it ties together my interests in AI and embedded systems while also offering a practical and engaging project scope. It is good to mention, while developing the project, I will continue with the machine learning specialization course, as I want to further strengthen my technical baseline for AI and ensure that I have the necessary skills to successfully execute the project.